DATA ANALYTICS FINAL PROJECT

1,440 ship trips. One surprising fuel finding.

Ship Fuel Efficiency & Emissions Analysis

Python | pandas | Matplotlib | Seaborn

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THE QUESTION

What drives fuel use and CO_2 on the water?

1. Which ship types and routes burn the most fuel and emit the most CO_2?

2. How much does weather change fuel use and engine efficiency?

3. Does fuel type (HFO vs Diesel) change efficiency or emissions?

Useful for route planning, fuel budgeting and maintenance priorities.

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THE DATA AND THE METHOD

From raw trips to clean insight

1,440
trip records

10
features

4 + 4
ship types + routes

0
missing values

0
duplicate rows

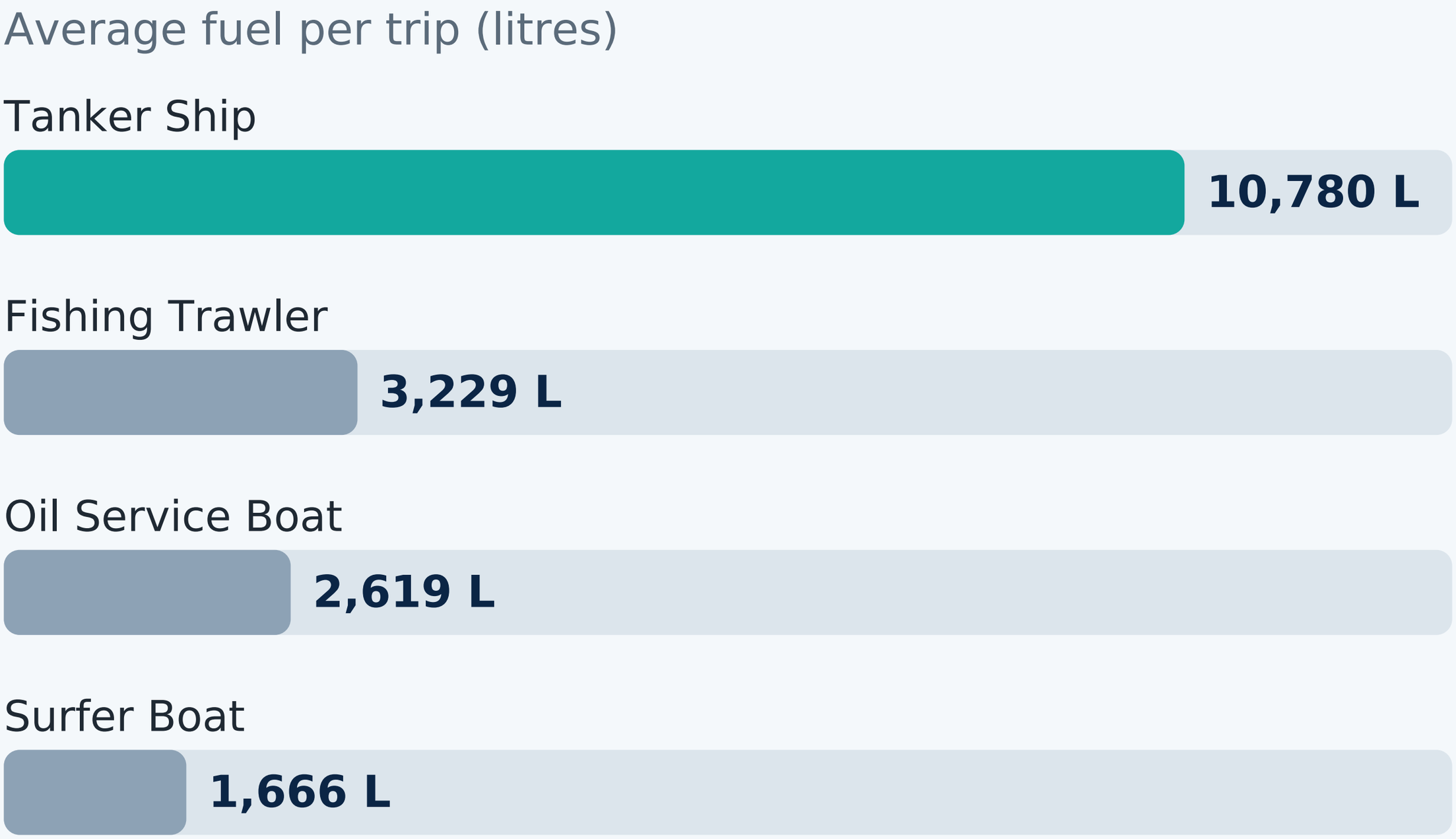
4
engineered features

Features built: distance per litre, CO_2 per litre, CO_2 per km, season. Then **16 visualisations**: univariate, bivariate and multivariate.

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FINDING 1

Tankers burn 6.5x the fuel of a Surfer Boat



Average CO_2 per trip: **Tanker 29,795 kg** vs **Surfer Boat 4,578 kg**. Tankers also travel the furthest (~267 km on average).

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FINDING 2

CO_2 follows fuel almost perfectly

	Distance	Fuel	CO_2	Efficiency
Distance	1.00	0.95	0.94	-0.03
Fuel	0.95	1.00	0.997	-0.03
CO_2	0.94	0.997	1.00	-0.03
Efficiency	-0.03	-0.03	-0.03	1.00

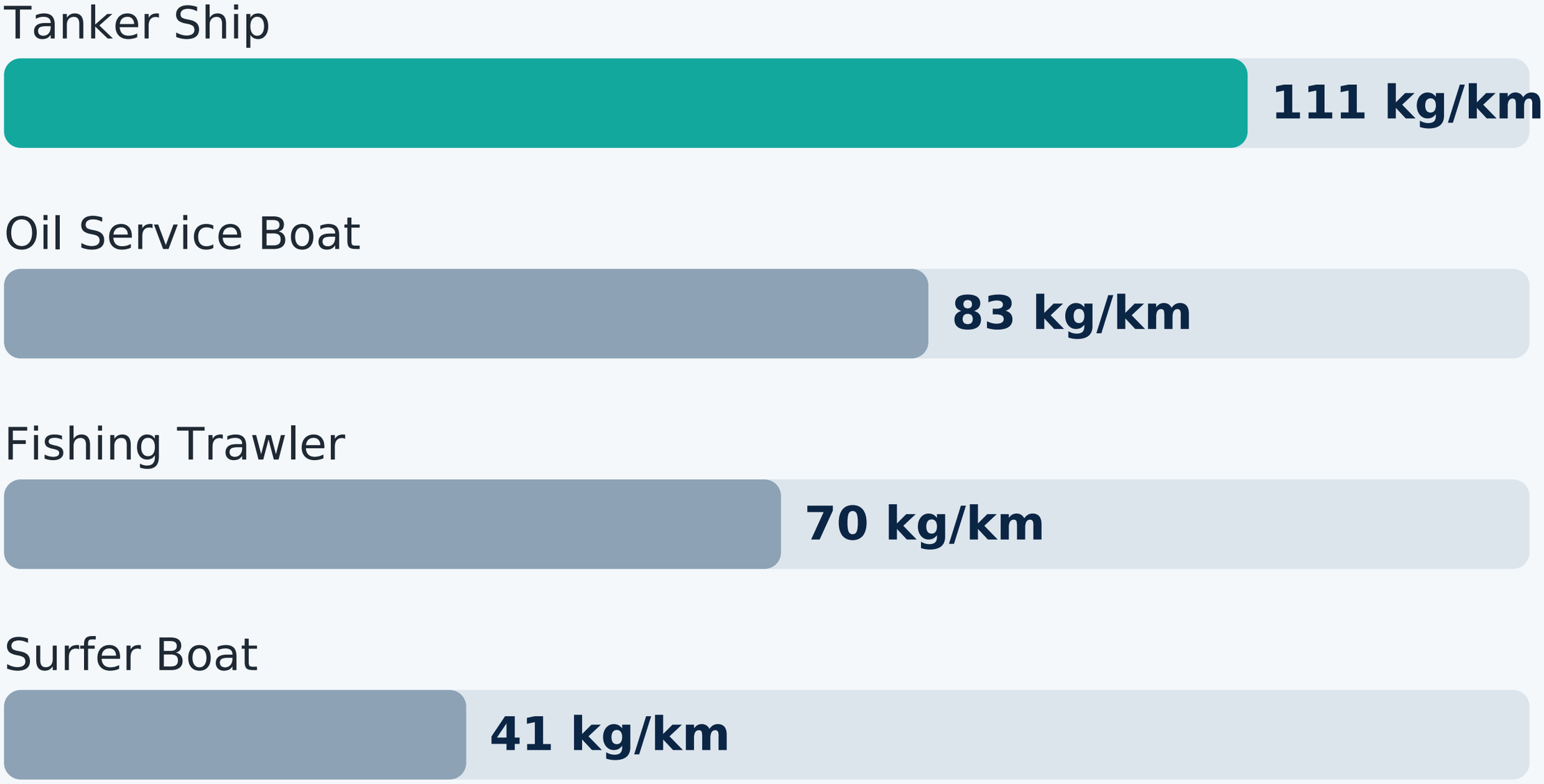
r = 0.997 between fuel and CO_2. About **2.75 kg of CO_2 per litre** on every vessel. Engine efficiency barely correlates with anything (r ≈ -0.03).

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FINDING 3

The real lever: CO_2 per km

Average CO_2 emitted per km travelled (kg)

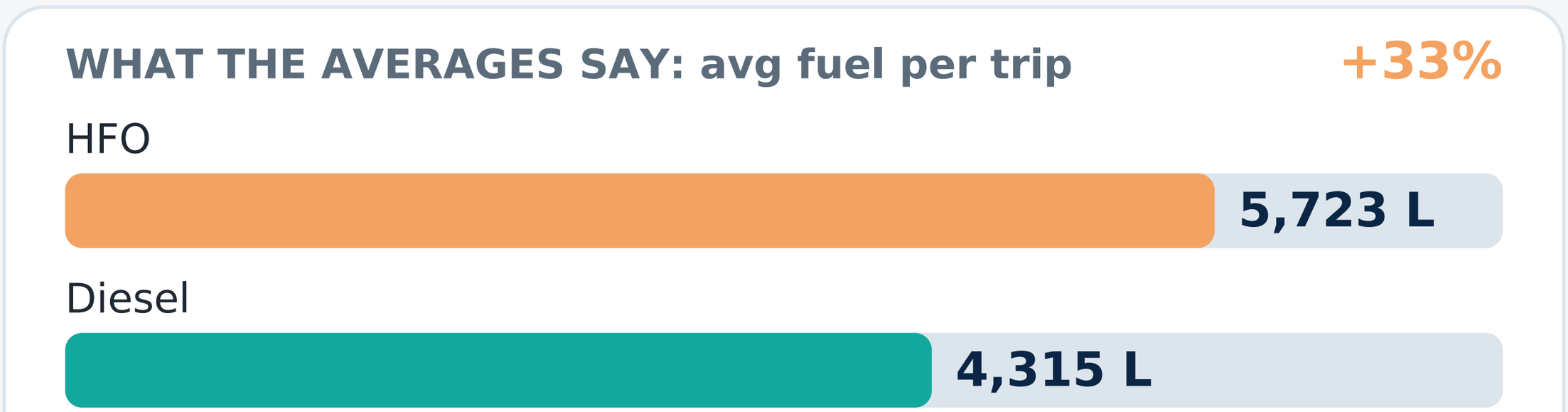


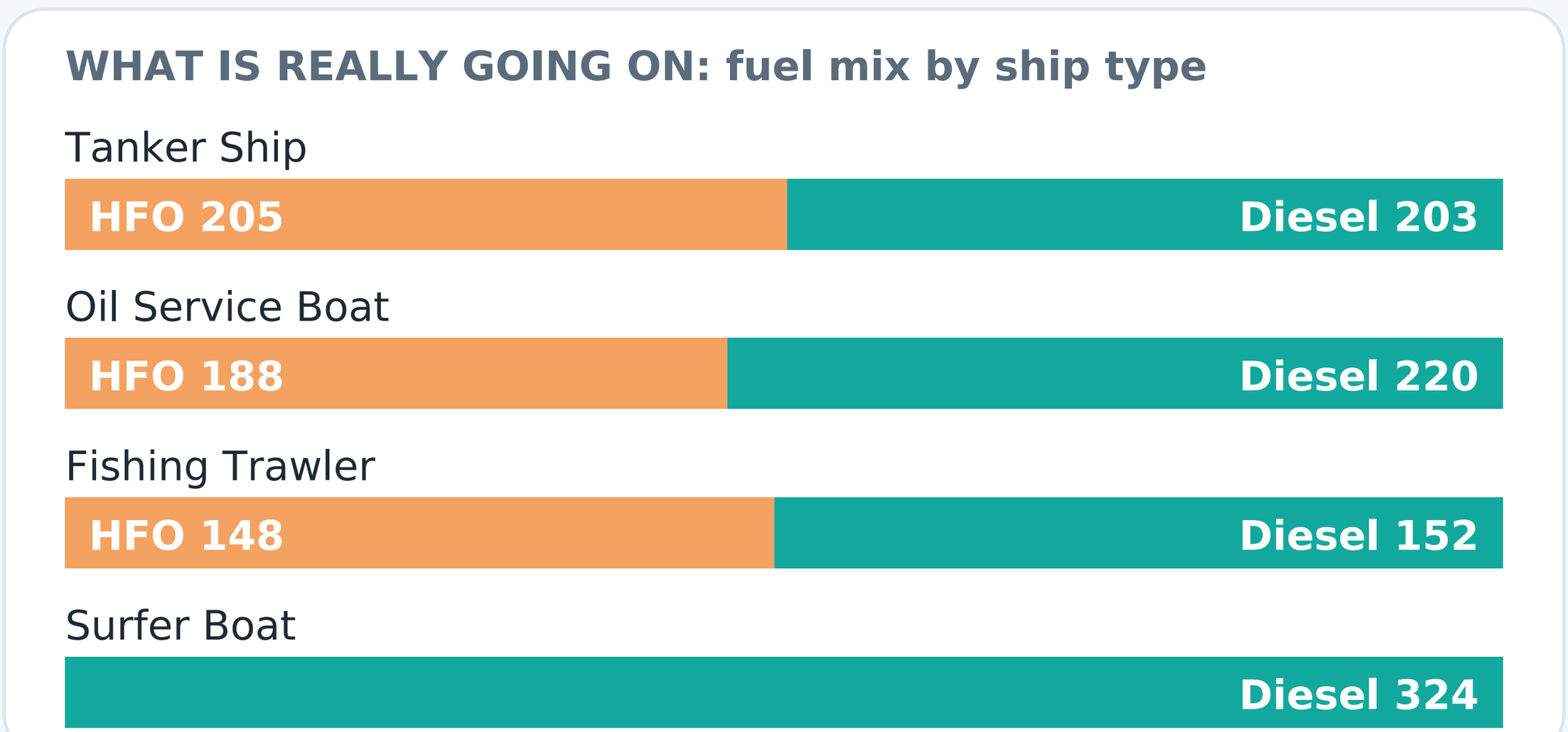
A **~3x gap** between ship types, while CO_2 per litre barely changes. Ship and route choice matter more than fuel alone.

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FINDING 4

The HFO “penalty” is an illusion



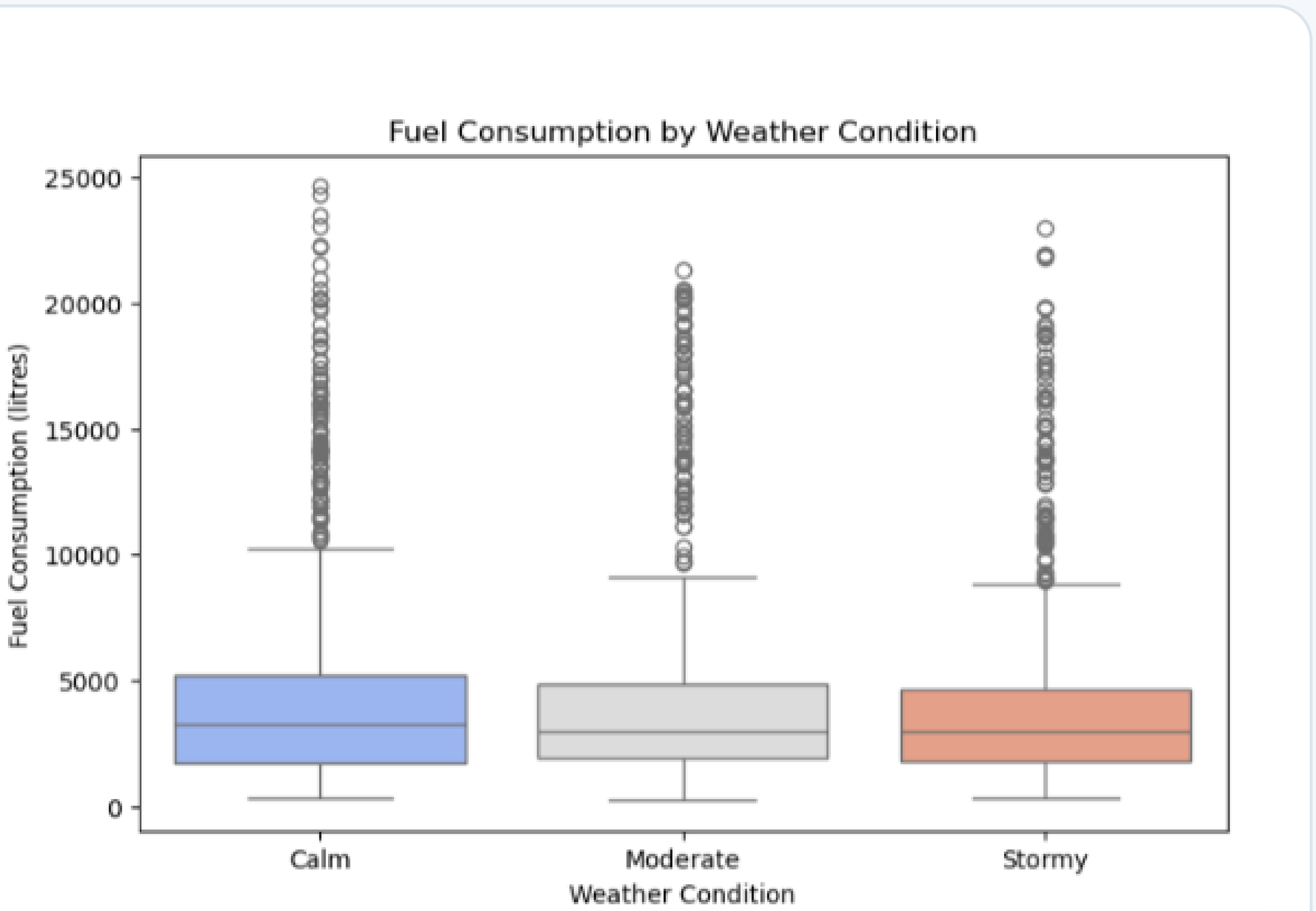


Surfer Boats, the lightest fuel users, never run on HFO. **Compare fuels within a ship type.**

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FINDING 5

Weather barely moved the needle



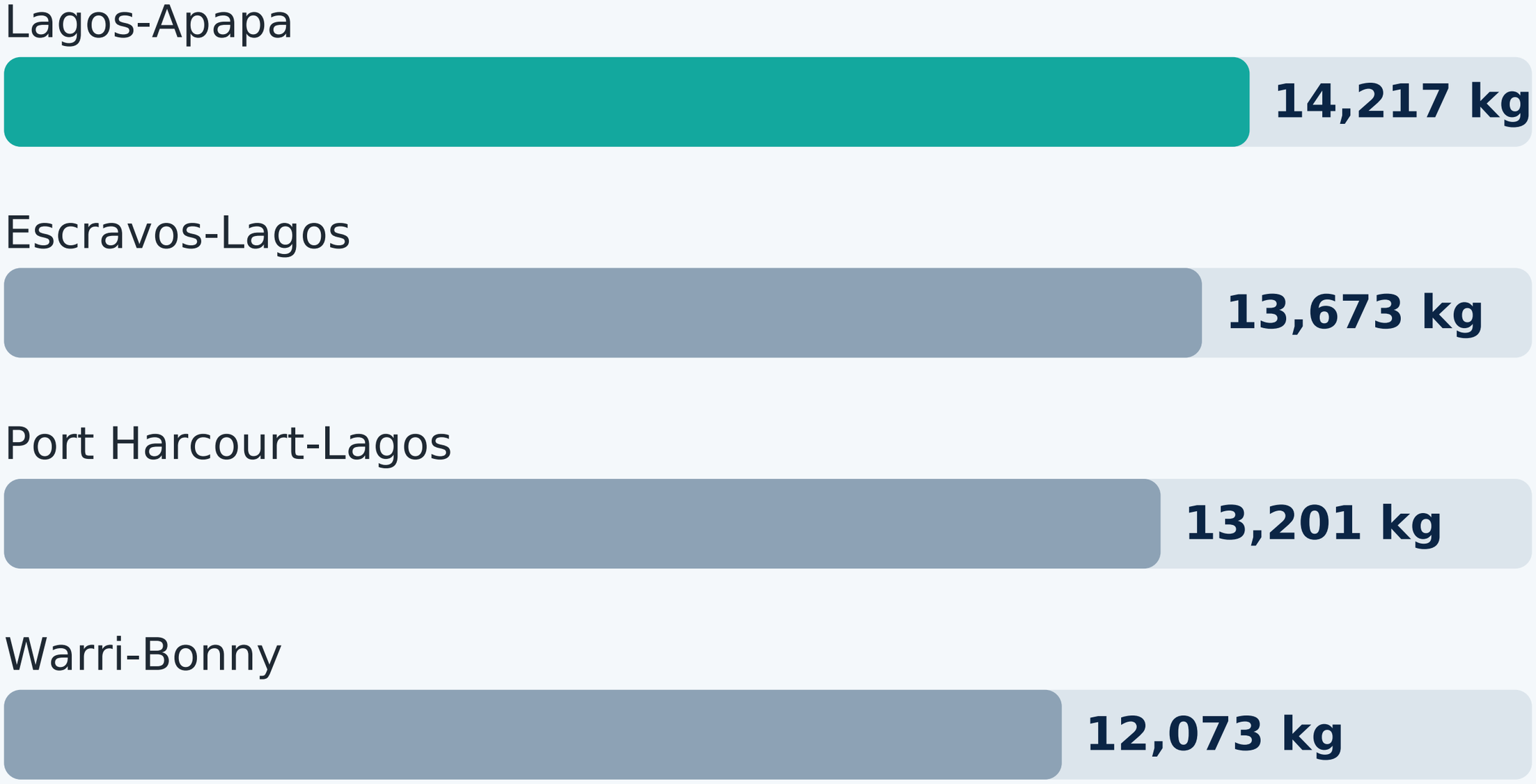
Calm, Moderate and Stormy trips look alike on **fuel use and engine efficiency** in this dataset. Ship type and route mattered far more.

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FINDING 6

Where average emissions are highest

Average CO_2 per trip by route (kg)



Lagos-Apapa is highest on average, narrowly ahead of Escravos-Lagos. **Warri-Bonny** is lowest. Useful context for route-level planning.

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WHAT I WOULD DO NEXT

From insight to action

1

Audit tankers and Lagos-Apapa first

Largest absolute savings potential.

2

Compare fuels within one ship type

Removes the ship-assignment effect.

3

Judge fuel switching on cost and emissions

Fuel type showed little direct efficiency effect.

4

Find what drives engine efficiency

Vessel age, maintenance and engine type are not in this data.

Limits: one dataset, so treat findings as a demonstration of method rather than fleet-wide facts.

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LET’S TALK

What would you check next?

Full code and notebook on GitHub (link in the first comment). Maritime and data folks, I would value your feedback.

Scan for the repo

github.com/Aseemkaipakasseri1994/ShipFuel-Analytics-Suite

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